

Sidharth

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Ann Arbor, Michigan - 48105, USA

Research Interests

Speech & Audio AI · Speech Enhancement & Extraction · Representation Learning · Spatial Audio

Education

- University of Michigan August 2025 - May 2029
PhD in Computer Science and Engineering Ann Arbor, MI, USA
 - Advisor: [Dr. Dhruv “DJ” Jain](#)
 - Collaborator: [Dr. Hao-Wen \(Herman\) Dong](#)
 - Research Focus: Audio AI, Speech Representation Learning
- University of Washington September 2023 - June 2025
MS in Electrical Engineering Seattle, WA, USA
 - Advisors: [Dr. Rajesh Rao](#), [Dr. Jeffrey Herron](#)
 - Research Focus: Artificial Intelligence, Signal Processing.
 - Thesis: "Decoding Pain: Statistical Identification of Biomarkers from Electrophysiological Signals"
- College of Engineering Trivandrum July 2019 - August 2023
B.Tech in Electronics and Instrumentation Engineering with minor in Mathematics Trivandrum, KL, INDIA
 - Advisor: [Dr. Jerrin Thomas Panachakel](#)
 - Thesis: "Emotion Detection from EEG using Transfer Learning"

Industrial Research Experience

- Bose Research  June 2026 - August 2026
Research Intern, Spatial Audio / Machine Learning Framingham, MA, USA
Advisor: Li-Chia (Richard) Yang, Dr. Mikolaj Kegler, Dr. Shuo Zhang
 - Developed an queryable open-vocabulary SELD system for query-based localization of arbitrary sound events in multichannel audio; identified absent-query false localizations as a key reliability failure mode.
 - Proposed negative-query representation learning + query verification, reducing query presence/absence EER from 19.36% to 12.7% while preserving standard SELD performance on TNSSE21/STARSS23.
- Skyworks Solutions.  May 2025 - August 2025
Research Intern Hillsboro, OR, USA
Advisor: Dr. Meysam Asgari
 - Pioneered a dual-microphone framework fusing in-ear and external signals to boost intelligibility in low-SNR conditions
 - Designed a low-parameter (1.7M), streamable U-Net with cross-attention for real-time two-channel enhancement.
 - Built a real-world dataset using a dummy-head HAT system for robust ecological evaluation.
 - Explored a BERT-based metric to quantify speech-drop effects.
 - Achieved DNSMOS OVRL of 2.45 on real-world collected data (very low SNR).
- BrainChip Research  June 2024 - March 2025
Research Intern Laguna Hills, CA, USA
Advisor: Dr. Yan Ru Pei
 - Co-led development of aTENNuate, a real-time deep state-space speech enhancement model
 - Achieved PESQ 3.27 on VB-DMD dataset with only 0.84M parameters, 0.33G MACs, and 46.5 ms latency.
 - Advanced deployment-specific optimization of SSMs via LoRA adaptation, enabling fine-tuning for diverse environments.
 - Led the development of SSM-based TTS system that achieved a MOS of 3.84.

Academic Research Experience

- Soundability Lab.  August 2025 - Present
Graduate Student Research Assistant
Ann Arbor, MI, USA
Advisor: Dr. Dhruv Jain, Dr. Herman Dong
 - Built an enrollment-free target speech extraction system that predicts speaker embeddings directly from a noisy mixture, removing the need for clean reference audio and simplifying deployment in real-world multi-speaker settings
 - Developed a WavLM-based teacher-student pipeline with permutation-invariant training to produce robust multi-speaker identity embeddings; the 2-speaker model reached 75.52% clustering accuracy, 0.72 NMI, and 0.58 ARI, far outperforming a WavLM + K-means baseline (7.35% accuracy) on overlapped noisy mixtures
 - Integrated the embeddings into production-relevant speech models and achieved up to 12.83 dB SI-SDRi on 2-speaker mixtures and 11.65 dB SI-SDRi on 3-speaker mixtures with DPCCN, while also validating generalization on 641 real DNS Challenge clips
- Mobile Intelligence Lab  March 2024 - June 2024
Graduate Student Researcher
Seattle, WA, USA
Advisor: Dr. Shyam Gollakota
 - Designed and implemented a model to extract conversation mixtures with volume-controlled target speaker output, aimed at improving flexibility and user focus in hearing aid applications.
 - Developed a d-vector module to generate speaker-specific embeddings for conditioning the separation model.
 - Enhanced the TF-GridNet architecture by integrating target speaker embeddings using techniques such as Feature-wise Linear Modulation (FiLM) for TSE
 - Achieved a PESQ score of 2.7 on the LibriMix speech separation dataset, demonstrating competitive performance.

Patents

- [P.1] Dhruv Jain, FNU Sidharth "Multi-Target Speech Extraction, Embedding and Enhancement (Pending 64/026,849)" U.S Provisional Patent filed from University of Michigan,
- [P.2] Y.R.Pei, R. Shrivastava, FNU Sidharth. [Real-time Speech Enhancement on Raw Signals with Deep State-space Modeling](#). U.S Patent filed from BrainChip, Inc
- [P.3] M Anthony Lewis, Amir Mohammad NADERI, Dr. Olivier Jean-Marie Dominique COENEN, PhD, Dr. Yan Ru PEI, Kristofor D. CARLSON, FNU Sidharth, Ritik Shrivastava. ["Methods and Systems for Radar Signal Processing Using Temporal Neural Networks with State-Space Model"](#). U.S Patent filed from BrainChip, Inc
- [P.4] Dhruv Jain, Jeremy Huang, FNU Sidharth, "Context-Aware Sound Selection and Suppression for Noise-Sensitive People (Pending, 64/101,380.)" U.S Provisional Patent filed from University of Michigan.

Publications

C=Conference, W=Workshop, J=Journal, S=In Submission, T=Thesis

- [S.1] FNU Sidharth, Meysam Asgari, Hao-Wen Dong, Dhruv Jain "Unmixing The Crowd: Learning Persistent Speaker Representations from Mixture-Derived Multi-Speaker Embeddings"
- [S.2] Jeremy Huang, Emani Hicks, Sidharth, Gillian Hayes, Dhruv Jain "Sona: Real-Time Multi-Target Sound Attenuation for Noise Sensitivity".
- [C.1] Y.R.Pei, R. Shrivastava, FNU Sidharth "Optimized Real-time Speech Enhancement with Deep SSMs on Raw Audio". In Proc. Interspeech 2025, pp. 51-55. DOI: 10.21437/Interspeech.2025-19
- [W.1] Sidharth, Vishwas Sathish, et.al. "PainDECOG: Machine Learning-Based Identification of Pain Biomarkers from sEEG Signals". In 2025 AAAI Workshop on Health Intelligence (W3PHIAI-25)
- [C.2] Baghel, S., Ramoji, S., Sidharth "The DISPLACE Challenge 2023 - Diarization of SPeaker and LAnguage in Conversational Environments". In Proc. Interspeech 2023, pp. 3562-3566. DOI: 10.21437/Interspeech.2023-2367
- [C.3] Sidharth, et.al. "Emotion Detection from EEG using Transfer Learning". In 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), pp. 1-4, DOI: 10.1109/EMBC40787.2023.10340389
- [C.4] J. T. Panachakel, R. H, S. P. K, S., Sidharth, et.al. "CSP- LSTM Based Emotion Recognition from EEG Signals". In 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRaine), pp. 289-294, DOI: 10.1109/MetroXRaine58569.2023.10405666
- [C.5] K. Sana Parveen, J. T. Panachakel, H. Ranjana, Sidharth, et.al. "EEG-based Emotion Classification - A Theoretical Perusal of Deep Learning Methods". In 2023 2nd International Conference for Innovation in Technology (INOCON), pp. 1-6, DOI: 10.1109/INOCON57975.2023.10101002

Relevant Projects

- **Masked Audio Modeling for Multi-Microphone Speech Reconstruction and Localization** March 2024 - May 2024
Tools: Numpy, PyTorch, Pyroomacoustics 
 - Developed a conformer-based masked audio modeling system for reconstructing missing microphone channels.
 - Simulated 4-mic room acoustics with pyroomacoustics, achieving validation MSE of 0.001
 - Built a data pipeline using PyTorch for diverse noisy environments to improve model robustness.
 - Adapted conformers to learn temporal-spatial correlations for source localization..
- **PainDECOG: Decoding Acute Pain from Intracranial EEG** March 2024 - September 2024
Tools: Python, NumPy, SciPy, Scikit-learn, MNE-Python, Matplotlib 
 - Developed machine learning models to classify acute pain states from raw iEEG signals using PIB and coherence features
 - Implemented preprocessing pipeline with notch/bandpass filtering, electrode selection, and 10-sec trial windowing.
 - Created dataset from multi-day iEEG recordings of 3 subjects with synchronized pain ratings for supervised learning.
 - Achieved up to 73% accuracy (RF, subject 3) in binary pain classification, outperforming chance levels.

Skills

- **Programming Languages:** Python, C++, MATLAB, Bash
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Hugging Face Transformers, Pyroomacoustics, gpuRIR
- **DevOps & Version Control:** Git, GitHub Actions, Slurm
- **Other Tools & Technologies:** Librosa, Praat, Triton Inference Server, LaTeX

Honors and Awards

- **NSF AAI, ECE DEI, and Weil Neurohub Travel Grant** Feb 2025
National Science Foundation (NSF), ECE/CSE Departments, University of Washington
 - Awarded \$2800 to present research paper "Decoding Pain: Statistical Identification of Biomarkers from Electrophysiological Signals" at AAAI 2025 Workshop on Health Intelligence, Philadelphia, USA.
 - Recognized for contributions at the intersection of AI, neuroscience, and health.
- **Travel Grant** Dec 2023
College of Engineering, Trivandrum
 - Awarded \$500 to present research paper "Emotion Detection from EEG using Transfer Learning" at an international conference in Sydney.
 - Supported for representing institution at a global research venue.
- **Winter Research Fellowship** Jan 2022
Indian Institute of Science (IISc), Bangalore
 - Awarded \$715 to conduct research on speaker and language diarization in multilingual Indian languages.
 - Enabled early independent research exposure in speech and language processing.

Service and Outreach

- **Admissions Committee Member** 2025
ECE MS Admissions Committee, University of Washington
 - Contributed to the review and selection process for incoming MS students.
 - Promoted fairness, diversity, and excellence in graduate admissions.
 - Gained insights into academic evaluation and admissions policies.
- **Founder** 2021
MATHLETES CET, College of Engineering Trivandrum, India
 - Established the official math club to encourage problem-solving and academic collaboration.
 - Organized student-led talks, quizzes, and workshops on advanced mathematics topics.
 - Built a platform fostering peer learning, leadership, and community engagement.

References

1. Dr. Dhruv Jain
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2. Dr. Hao-Wen (Herman) Dong
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3. Richard Yang
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4. Dr. Mikolaj Kegler
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